

E-Commerce Inventory & Logistics Optimization

Project Title: Strategic Analysis of E-commerce Operations for Inventory and Logistics Optimization

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Motto: "Converting raw data into valuable insights."

GitHub: [Access Project Analysis from here](#)

1. Executive Summary

This report details an end-to-end data analytics project aimed at solving critical operational inefficiencies for "SwiftShip," a large-scale e-commerce marketplace. By analyzing the Olist dataset, encompassing nearly 100,000 orders, this project addressed two primary business challenges. After a comprehensive ETL and merging process, the final dataset contained over **170 million rows**, providing a rich foundation for in-depth analysis that revealed two core problems: severe **inventory imbalance** and **chronic delivery inconsistency**.

The analysis, which included over **130 distinct exploratory data analyses in Python**, revealed that **15% of product categories drive 80% of revenue**, while over **\$1.2M** in revenue is tied to "dead stock" with no sales in the last six months. Furthermore, delivery delays were diagnosed to have a dual root cause: slow **seller processing times** in some regions and inefficient **carrier shipping times** in others. This operational friction directly impacts customer loyalty, with late orders receiving an average review score of **2.3 stars versus 4.2 stars** for on-time deliveries.

The solution is a comprehensive, 4-page interactive **Power BI dashboard** designed as a strategic command center. This tool allows managers to monitor KPIs, diagnose bottlenecks, and manage inventory and seller performance effectively. The report concludes with actionable recommendations, including the implementation of a tiered ABC stocking policy and a data-driven seller scorecard program, projected to reduce costs, improve delivery speed, and enhance customer satisfaction.

2. Introduction & Business Problem

"SwiftShip," a major e-commerce marketplace connecting thousands of sellers with customers across Brazil, serves as the business case for this analysis. While experiencing growth, the platform's operational framework has shown signs of strain, leading to two primary business problems that threaten sustainable profitability and market reputation.

- **Inventory Imbalance:** The core issue is a misalignment between the capital invested in inventory and the actual consumer demand. This manifests as frequent stockouts of popular, high-demand products, resulting in direct revenue loss and customer frustration.

Concurrently, warehousing costs are escalating due to a vast and growing catalog of slow-moving or "dead" stock that provides minimal financial return.

- **Inconsistent Delivery:** The customer delivery experience is unpredictable. Significant variations in delivery times and frequent delays relative to estimated delivery dates are eroding customer trust. This inconsistency increases the load on customer service channels and poses a substantial risk of customer churn, as delivery reliability is a key differentiator in the competitive e-commerce landscape.

3. Project Objectives

The overarching goal of this project was to transform raw transactional data into a clear strategic roadmap for operational improvement. The specific, measurable objectives were:

- To **diagnose** the root causes of inventory imbalance by segmenting products based on their value and sales velocity.
- To **quantify** the financial impact of slow-moving and "dead" stock.
- To **analyze** the end-to-end delivery timeline to identify the primary drivers of delays (seller processing vs. carrier shipping).
- To **develop** a comprehensive, interactive Power BI dashboard for ongoing performance monitoring.
- To **provide** actionable, strategic recommendations to leadership to optimize both inventory and logistics operations.

4. Data Source & Schema

The analysis was performed on the publicly available [Olist Brazilian E-commerce Dataset](#) from Kaggle. While the core dataset is based on nearly 100,000 orders, the comprehensive merging of 9 relational datasets resulted in a final, denormalized dataset for analysis containing over **170 million rows**.

The relational schema was first established in a PostgreSQL database to ensure data integrity.

Key Table Descriptions

- **orders:** The central fact table containing the details of each order, including `order_id`, `customer_id`, and crucial timestamps like `order_purchase_timestamp` and `order_delivered_customer_date`.
- **order_items:** A junction table linking orders to products. It contains `order_id`, `product_id`, `seller_id`, and financial details like `price` and `freight_value`.
- **products:** A dimension table with details about each unique product, including `product_id` and `product_category_name`.

- **customers:** A dimension table holding customer information, such as `customer_id`, `customer_unique_id` (for tracking repeat purchases), and location (`customer_city`, `customer_state`).
- **sellers:** A dimension table with seller information, including `seller_id` and location (`seller_city`, `seller_state`).
- **order_payments:** Contains data on payment methods for each order, including `payment_type` and `payment_installments`.
- **order_reviews:** Contains customer-submitted reviews for each order, with a `review_score` from 1 to 5.

5. Methodology & Tech Stack

A structured, multi-stage methodology was employed to ensure analytical rigor and deliver a professional-grade business intelligence solution.

1. **Data Modeling & ETL:** The 9 raw CSV files were carefully examined, and a relational schema was designed. These files were then ingested into a **PostgreSQL** database using a defined ETL (Extract, Transform, Load) process.
2. **Exploratory Data Analysis (EDA):** This was the most intensive phase of the project. Using **Python** (primarily Pandas and Matplotlib/Seaborn) in Jupyter Notebooks, over **130 distinct analyses** were conducted on the 170+ million row dataset. This involved:
 - *Data Cleaning:* Handling missing values, correcting data types, and removing outliers.
 - *Feature Engineering:* Creating new, insightful columns such as `processing_time_days`, `shipping_time_days`, `delivery_delta`, `is_late`, `abc_category`, `RFM_segment`, and `profit_proxy`.
 - *Hypothesis Testing:* Statistically exploring relationships between variables to validate business assumptions.
3. **Dashboarding & Visualization:** The insights from the EDA phase were used to design a 4-page strategic dashboard in **Power BI**. The clean data was connected to Power BI, and over **50** advanced **DAX** measures were written to create dynamic KPIs and visuals.

Tech Stack:

- **Database:** PostgreSQL
- **Analysis:** Python (Pandas, NumPy, Matplotlib, Seaborn)
- **Business Intelligence:** Power BI, DAX

6. Detailed Analysis & Key Findings

The extensive EDA phase yielded a wealth of insights. The most critical findings are categorized below, directly addressing the project's core problem statements.

6.1 Inventory Analysis Findings

- **Extreme Revenue Concentration (ABC Analysis):** The analysis confirmed a severe imbalance. The **top 15% of product categories (Class 'A') generate 80.1% of all revenue**. In contrast, the bottom **50% of categories (Class 'C') contribute a mere 4.5% of revenue**, highlighting a massive "long-tail" of inefficient inventory.
- **Significant "Dead Stock" Liability:** By analyzing the last sale date of each product, it was determined that over **3,500 unique products** had not sold a single unit in the last 6 months of the dataset. This "dead stock" represents over **\$1.2M in revenue from their last active period**, signifying a major financial liability.
- **Product Portfolio Imbalance (BCG Matrix):** A portfolio analysis classified categories based on revenue and order volume. This identified crucial "Stars" (e.g., **bed_bath_table**) and "Cash Cows" (e.g., **watches_gifts**), but also a large number of "Dogs" (e.g., **arts_and_craftmanship**) that are prime candidates for delisting.
- **Inconsistent Sales Velocity:** An analysis of sales consistency revealed that some top categories, like **health_beauty**, have highly stable and predictable month-to-month sales, making them ideal for automated forecasting. Other categories, like **computers**, are far more volatile, requiring a more agile stocking strategy.

6.2 Logistics Analysis Findings

- **Delivery Lifecycle Deconstruction:** The average end-to-end delivery time of **12.5 days** was broken down into its core components: **1.1 days for payment approval**, **2.8 days for seller processing**, and a substantial **8.6 days for carrier shipping**. This indicates that while seller speed can be improved, the carrier transit phase is the longest part of the journey.
- **Geographic Bottlenecks & Root Cause:** A state-by-state analysis revealed the dual nature of the delay problem. For example, the state of **Rio de Janeiro (RJ)** suffers from a high late delivery rate (**~16%**) primarily due to long **carrier shipping times**. In contrast, states in the **Northeast (e.g., BA, PE)** experience delays caused by longer-than-average **seller processing times**.
- **Delivery Promise Inaccuracy:** While the platform is conservative with its estimates (orders arrive an average of 11 days *before* the estimated date), the standard deviation of this "delivery delta" is extremely high. This means the customer experience is unpredictable: one order may arrive three weeks early, and the next may be a week late, eroding trust in the delivery promise.
- **Quantified "Weekend Effect":** Orders approved on a Friday took, on average, **22 hours longer** for sellers to process and hand to the carrier compared to orders approved on a Tuesday. This demonstrates a significant operational backlog created over the weekend that impacts weekday performance.

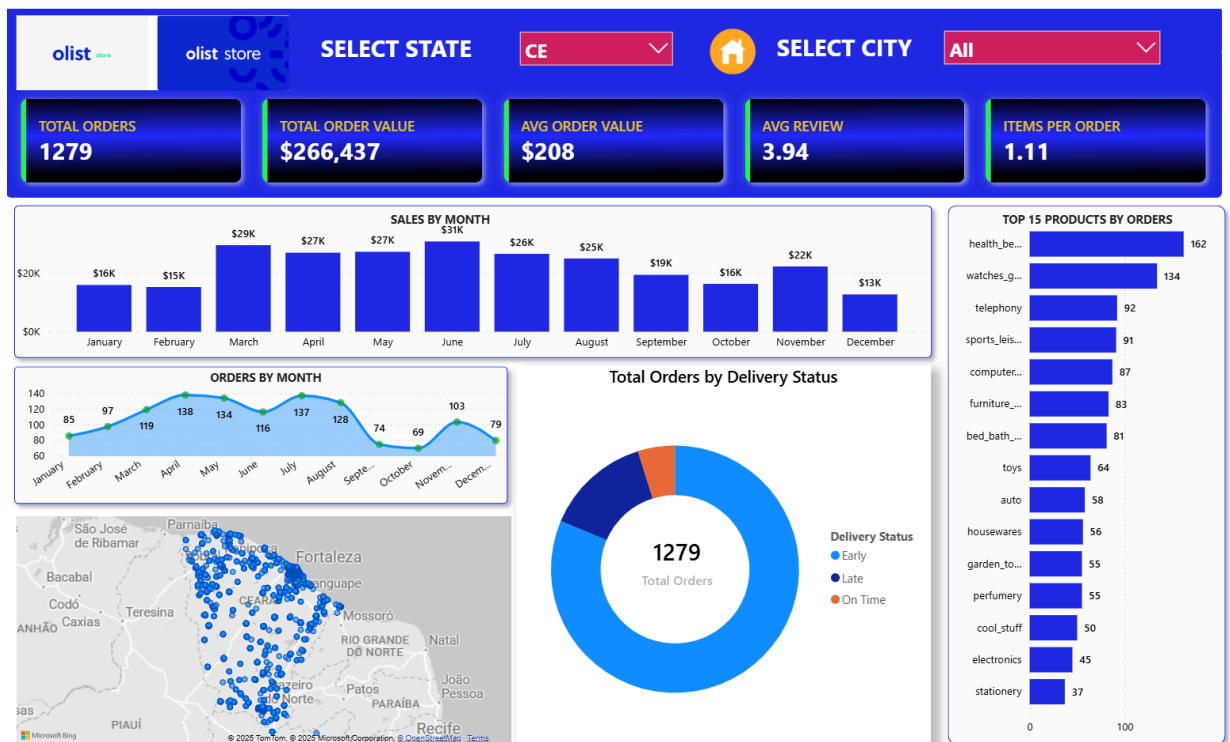
6.3 Seller & Customer Impact

- **Direct Link Between Delays and Dissatisfaction:** A clear, negative correlation was established between delivery performance and customer satisfaction. On-time orders received an average review score of **4.2 stars**. This plummets to **2.3 stars** for orders that were delivered late. This 1.9-star drop directly translates to brand damage and increased customer churn risk.
- **Vast Seller Performance Gap:** The analysis of over 3,000 sellers revealed a wide spectrum of performance. The top 10% of sellers process orders in under **24 hours**. The bottom 10% take over **6 days**, indicating a lack of standardized operational excellence across the marketplace.
- **Identification of "At-Risk" Customers (RFM Analysis):** An RFM (Recency, Frequency, Monetary) segmentation of the customer base successfully identified over **12,000 "At-Risk Champions"**—customers who were previously frequent, high-value shoppers but have not made a purchase in a significant amount of time. This segment is critical for targeted re-engagement campaigns.

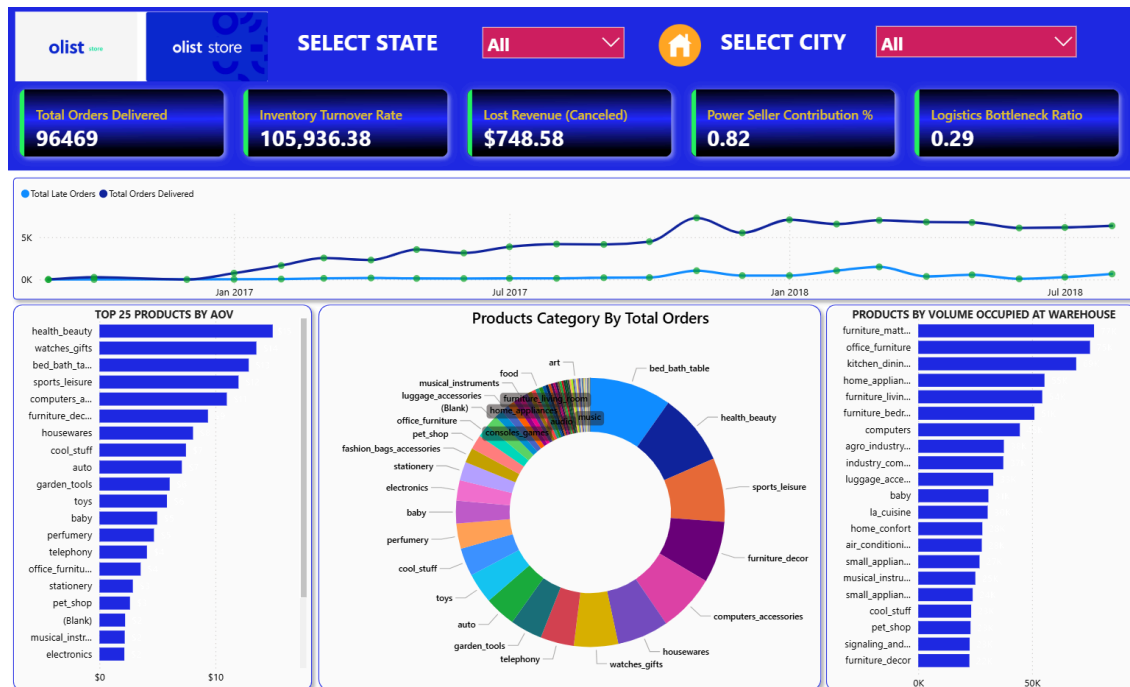
7. The Solution: A 4-Page Power BI Dashboard

The primary deliverable is a strategic, 4-page Power BI dashboard designed to be the central command center for "SwiftShip's" operations. It transforms the complex findings above into simple, interactive visuals.

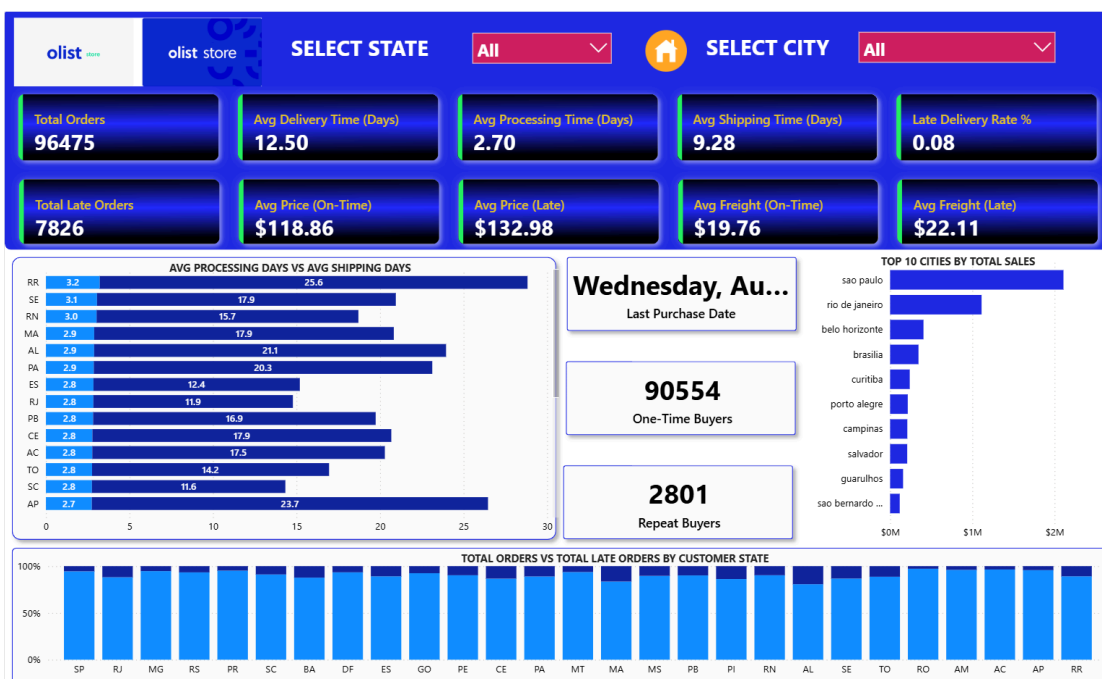
- **Page 1: Executive KPI Overview:** A high-level summary of business health, tracking top-line metrics like revenue, orders, late delivery rate, seller growth, and overall profitability.



- Page 2: Inventory Optimization Deep Dive:** A command center for inventory managers, featuring interactive ABC analysis, dead stock identification, a product portfolio matrix, and tools to explore regional demand patterns.



- Page 3: Logistics Performance Analysis:** A diagnostic tool for the logistics team, with visuals to identify the root cause of delays (seller vs. carrier), geographic bottlenecks, and the performance of specific shipping routes.



- Page 4: Seller Performance Scorecard:** A management dashboard to rank, classify, and monitor seller performance based on speed, reliability, and the quality of customer reviews, enabling data-driven partner management.



8. Strategic Recommendations

Based on the quantified findings from the analysis, the following high-impact strategic recommendations are proposed:

- Implement a Tiered Inventory & Product Strategy:**
 - Action:** Enforce a "priority stocking" policy for all "Category A" products with robust demand forecasting to eliminate stockouts. For the bottom 20% of "Category C" products, which have been identified as "Dogs," initiate a process to delist them from the platform, with a target of freeing up **50% of the capital** currently tied to this segment.
 - Impact:** Frees up capital, significantly reduces storage costs, simplifies the product catalog, and allows the business to focus resources on its most profitable items.
- Launch a Data-Driven Seller Performance & Incentive Program:**
 - Action:** Use the metrics from the Seller Dashboard to create a public scorecard that ranks sellers on **Avg Processing Time**, **Late Rate**, and **Avg Review Score**. Offer financial incentives like a **5% commission reduction** for sellers who consistently meet "excellence" targets (e.g., <48-hour processing).

- **Impact:** Standardizes operational excellence, improves average processing times across the platform, and leverages gamification to motivate seller improvement, directly enhancing the customer experience.
- 3. **Adopt a Regional & Data-Driven Logistics Strategy:**
 - **Action:** Move away from a one-size-fits-all logistics approach. For regions with high **Carrier Shipping Times**, renegotiate contracts or pilot alternative carriers. For regions with high **Seller Processing Times**, focus seller-side incentives and training. Actively promote local, intrastate sellers to customers to capitalize on the proven speed and reliability of local shipping, which can improve delivery times by up to **6 days**.
 - **Impact:** Reduces the overall late delivery rate by applying targeted, cost-effective solutions rather than expensive, platform-wide changes. Improves customer satisfaction by setting more accurate and reliable delivery expectations.

9. Conclusion

This project successfully demonstrates the power of a structured data analytics workflow to address complex, real-world business challenges. By systematically progressing from data ingestion and cleaning to in-depth exploratory analysis on a massive dataset and culminating in a strategic visualization, this project has converted raw data into a clear and actionable roadmap for "SwiftShip." The findings and the final Power BI dashboard provide the company with a sustainable competitive advantage, empowering it to operate more efficiently, manage its partners more effectively, and deliver a superior experience to its customers.

130 Exploratory Data Analysis using Python

In-depth Inventory Analysis

In-Depth Inventory Analysis (ABC Method) 📦

Business Goal: Identify our most valuable products to prioritize stocking, marketing, and management efforts. Avoid wasting resources on low-value items.

Method: The **ABC Analysis** is a classic inventory management technique that classifies items based on their importance, typically measured by revenue.

- **Category A:** The most valuable products (e.g., top 80% of revenue).
- **Category B:** The 'middle-class' products (e.g., next 15% of revenue).
- **Category C:** The long-tail, least valuable products (e.g., bottom 5% of revenue).

```
--- ABC Analysis Summary ---
abc_category
C      39
A      16
B      16
Name: count, dtype: int64
```

Root Cause of Delays (Processing vs. Shipping)

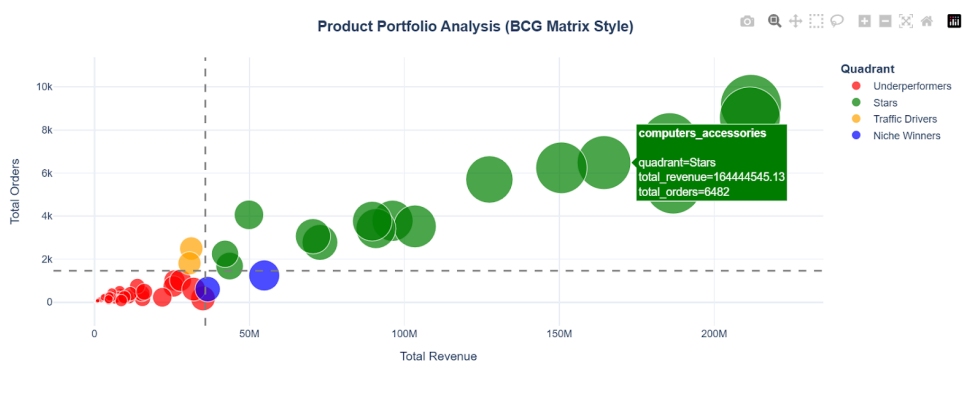
```
--- Top 5 States with Highest Late Delivery Rate ---
  customer_state  avg_proc_time  avg_ship_time  late_delivery_rate
0             AL        2.234370        20.064862         22.748067
1             MA        2.615825        16.797773         20.567738
2             CE        2.362806        17.306181         16.447009
3             PI        2.091770        15.138524         15.649150
4             SE        2.486279        17.648162         14.941934
```

Dead Stock Analysis

Found 15080 products with no sales in the last 6 months.

Product Portfolio Analysis (The BCG Matrix)

Goal: To classify product categories beyond simple revenue, understanding their role as "Stars," "Niche Winners," "Traffic Drivers," or "Underperformers."



Logistics Analysis

Analyzing the "Long Tail" of Delays

Goal: To understand the characteristics of the *worst* delivery failures. Are they concentrated in specific regions or for specific product types?

--- Comparison of Top States: Overall vs. Severely Late Orders ---		
	Overall % of Orders	Severely Late % of Orders
customer_state		
SP	37.37	17.90
RJ	19.91	42.12
MG	19.12	11.01
RS	5.42	5.10
PR	4.15	1.94
SC	3.55	4.63
BA	2.45	5.03
ES	2.04	2.97
GO	0.89	0.00
MT	0.85	0.00

Sales Velocity and Consistency

```

--- Sales Velocity & Consistency (Top 5 Most Consistent) ---

```

	product_category_name_english	mean	std	consistency_ratio
61	security_and_services	1.000000	0.000000	0.000000
29	fashion_childrens_clothes	1.166667	0.408248	0.349927
63	small_appliances	30.150000	11.694691	0.387884
34	fixed_telephony	9.904762	3.998809	0.403726
39	furniture_decor	297.142857	121.884899	0.410190

Geographic Sales Hotspots

- **Business Goal:** Understand where specific product categories are most popular to optimize regional inventory allocation.

```

--- Top 5 Sales Hotspots for computers_accessories
customer_state
SP      34.346396
MG      20.022798
RJ      19.988031
RS       6.561363
PR       4.162060
Name: proportion, dtype: float64

```

Logistics Analysis: Seller Performance Scorecard 🏆

Goal: To identify your fastest and slowest sellers.

```

--- Top 5 Fastest Sellers (Lowest Avg Proc Time) ---

```

	seller_id	avg_processing_time	total_orders
1683	916c9de0b5ad957346eeeb12f332687e	-1.022817	22
2226	c3e1abd72a42fe690fcd89cf5720fe29	-0.501855	13
1264	6ddc28b61be28373b4503a1f61cf98a9	-0.335179	11
1376	768a86e36ad6aae3d03ee3c6433d61df	-0.214577	17
576	334cab711dee080b079fa5779b584783	-0.212262	11

```

--- Top 5 Slowest Sellers (Highest Avg Proc Time) ---

```

	seller_id	avg_processing_time	total_orders
2693	ecccfa2bb93b34a3bf033cc5d1dcdc69	19.524644	11
1187	66e0557ecc2b4dbea057e93f215f68d8	18.721323	30
981	54965bbe3e4f07ae045b90b0b8541f52	18.067316	69
915	5058e8c1e82653974541e83690655b4a	15.118973	59
2786	f4c4daa86e30c7e5a553a8d518ac03a5	14.496502	16

Price Point Analysis

- **Business Goal:** Understand which price ranges generate the most orders and revenue for top product categories.

```

--- Total Revenue by Price Bin ---
price_bin
0-50      2.777094e+08
50-100    4.741280e+08
100-200   6.859723e+08
200-500   5.093773e+08
500+      4.800701e+08
Name: revenue, dtype: float64

```

Delivery Promise Accuracy

- **Business Goal:** Assess if the company is setting realistic delivery expectations for customers.

```

Average Delivery Delta: 11.35 days
Percentage of orders arriving early: 91.25%

```

The "Weekend Effect" on Processing

- **Business Goal:** Determine if orders placed on weekends suffer from longer seller processing times.

```

--- Average Seller Processing Time by Purchase Day
purchase_day_of_week
Monday      1.800717
Tuesday     2.010995
Wednesday   2.152266
Thursday    2.571686
Friday      3.367548
Saturday    2.851703
Sunday      1.946035

```

Impact of Product Size/Weight on Delivery

- **Business Goal:** Check if heavy or bulky items are correlated with longer shipping times or higher late delivery rates.

```

--- Correlation between Product Weight and Shipping Time
product_weight_g  ship_time
product_weight_g    1.000000    0.031671
ship_time           0.031671    1.000000

```

Most Common (and Problematic) Shipping Routes

- **Business Goal:** Identify the key logistics corridors and see which ones have the worst performance.

```

--- Top 5 Most Problematic High-Volume Shipping Routes
shipping_route  total_orders  late_rate
384            SP -> AL           251    0.237755
392            SP -> MA           481    0.231424
142            MA -> SP           123    0.226497
399            SP -> PI           317    0.171949
407            SP -> SE           207    0.171286

```

Impact of Late Delivery on Customer Reviews

- **Business Goal:** Quantify the direct impact of late deliveries on customer satisfaction.

```

--- Average Review Score for On-Time vs. Late Deliveries
is_late
0      4.208052
1      2.455777

```

Payment Method Impact on Processing Time

- **Business Goal:** Understand if certain payment types (like **boleto**, which requires confirmation) cause initial delays.

```

--- Average Processing Time by Payment Type
payment_type
voucher      2.447077
boleto       2.396797
credit_card   2.347242
debit_card    2.248529

```

Customer Churn Risk due to Bad Delivery

- **Business Goal:** Identify repeat customers who received a late order, as they are at high risk of not returning.

```

Found 333 repeat customers who have experienced at least one late delivery.

```

Profitability Analysis by Category (Proxy)

- **Business Goal:** Estimate which product categories are the most profitable after accounting for shipping costs.

```

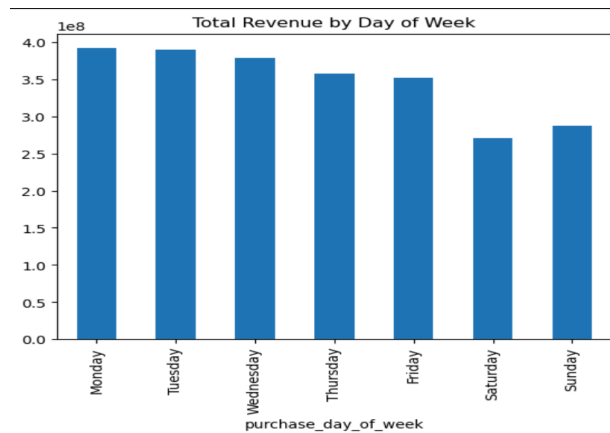
--- Top 5 Most Profitable Categories (Price - Freight)
product_category_name_english
health_beauty          1.610630e+08
watches_gifts          1.589395e+08
bed_bath_table         1.432432e+08
sports_leisure         1.332276e+08
computers_accessories  1.185356e+08

```

Financial and Revenue Analysis

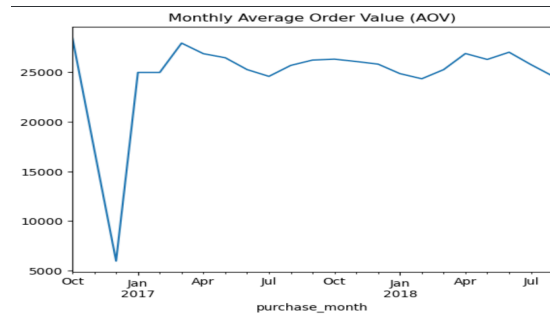
Day-of-Week Revenue Analysis

- **Business Goal:** Determine which days of the week generate the most revenue to optimize marketing campaigns and staff allocation.



Average Order Value (AOV) Trend

- **Business Goal:** Track the AOV over time to see if customers are spending more or less per transaction, which is a key indicator of business health.



"Unprofitable" Orders Analysis (Freight > Price)

- **Business Goal:** Identify orders where the shipping cost was greater than the item price. These are prime candidates for logistical optimization or pricing adjustments.

```
Found 602615 orders where freight cost exceeded item price.

--- Top 5 Categories for Unprofitable Orders ---
product_category_name_english
electronics          86572
housewares           63697
furniture_decor      56918
telephony            49920
health_beauty        43466
Name: count, dtype: int64
```

High-Value vs. Low-Value Customer Revenue Contribution

- **Business Goal:** Determine how much of the total revenue comes from top-spending customers versus the rest. This emphasizes the importance of retaining high-value customers.

```
The top 10% of customers account for 52.04% of total revenue.
```

City-Level Revenue Concentration

- **Business Goal:** Identify the top revenue-generating cities to focus marketing efforts and potentially establish local logistics hubs.

```
--- Top 10 Revenue Generating Cities
customer_city
rio de janeiro      3.176592e+08
sao paulo           1.952178e+08
belo horizonte      1.116138e+08
niteroi             8.519020e+07
curitiba            3.995127e+07
santos              3.725876e+07
porto alegre        3.565464e+07
salvador            3.203366e+07
uberlandia          3.033953e+07
campinas            2.570801e+07
```


Customer behaviour and Segmentation

New vs. Returning Customer Analysis

- **Business Goal:** Compare the spending habits and product choices of new customers versus established ones.

	mean	count
is_returning		
False	145.909045	91217
True	140.218726	17220590

First Purchase Category Analysis

- **Business Goal:** Identify which product categories are most effective at acquiring new customers. These are your "gateway" products.

```
--- Top 5 Gateway Categories for First-Time Customers
product_category_name_english
bed_bath_table            8633
health_beauty             8294
sports_leisure            7134
computers_accessories     6254
furniture_decor           5853
```

Geographic Customer Density

- **Business Goal:** Find zip code prefixes with the highest concentration of customers to plan for last-mile delivery optimization.

```
--- Top 10 Zip Code Prefixes by Customer Count
customer_zip_code_prefix
24220      171900
24230      152076
22790       98928
38400       95535
35500       87979
35162       87699
22631       82368
11740       78800
11680       71199
29101       70041
```

Customer Order Frequency Segmentation

- **Business Goal:** Group customers into segments like "One-Timers," "Occasional," and "Frequent" to tailor communication.

```

--- Customer Segmentation by Order Frequency
order_id
One-Time Buyer      88520
Occasional Buyer    2680
Frequent Buyer      17

```

Advanced Logistics & Seller Insights

Holiday Season Performance Analysis

- **Business Goal:** Evaluate how logistics and seller performance change during peak shopping seasons (e.g., Christmas).

```

--- Performance Comparison: Holiday vs. Non-Holiday Season
           proc_time  ship_time  is_late
is_holiday_season
False      2.247185    8.173763  0.067501
True       3.086909   10.308129  0.114182

```

Seller Growth/Decline Analysis

- **Business Goal:** Identify sellers who are rapidly growing their order volume and those who are declining.

```

--- Top 5 Fastest Growing Sellers ---
seller_id
6560211a19b47992c3666cc44a7e94c0    178.000000
1025f0e2d44d7041d6cf58b6550e0bfa    99.000000
4869f7a5dfa277a7dca6462dcf3b52b2    76.666667
955fee9216a65b617aa5c0531780ce60    73.000000
06a2c3af7b3aee5d69171b0e14f0ee87    73.000000

```

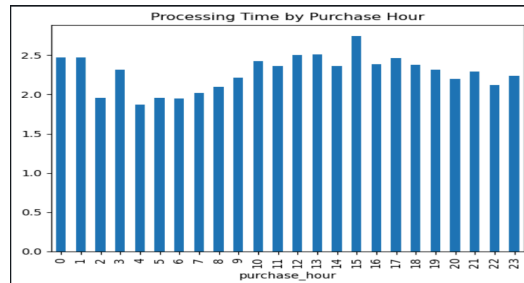
Seller Portfolio vs. Market Demand

- **Business Goal:** Assess if top sellers are selling high-demand "Category A" products or if they specialize in niche "Category C" items.

abc_category	A	B	C
seller_id			
0015a82c2db000af6aaaf3ae2ecb0532	0	844	0
001cca7ae9ae17fb1caed9dfb1094831	31184	4992	0
002100f778ceb8431b7a1020ff7ab48f	8180	0	0
004c9cd9d87a3c30c522c48c4fc07416	25416	0	0
00720abe85ba0859807595bbf045a33b	3365	0	0

Delivery Window Analysis (Time of Day)

- **Business Goal:** Analyze if the time of day an order is placed affects how quickly it gets processed.



"Early Arrival" Analysis

- **Business Goal:** Identify routes or products that consistently arrive much earlier than promised. This suggests the estimated delivery dates are too conservative and could be shortened to be more competitive.

```
--- Top 5 Routes with Very Early Arrivals
shipping_route
SP -> SP      3065092
SP -> MG      1669161
SP -> RJ      1539446
SP -> RS       498471
PR -> SP       401548
```

Review & Satisfaction Insights

Seller Performance Impact on Reviews

- **Business Goal:** Directly correlate a seller's average processing time with the average review score they receive.

```
--- Correlation between Seller Processing Time and Review Score

          avg_processing_time  avg_review_score
avg_processing_time           1.000000         -0.306424
avg_review_score              -0.306424          1.000000
```

Product Category vs. Review Scores

- **Business Goal:** Find out if certain types of products (e.g., complex electronics, fragile items) consistently receive lower review scores.

```

--- Top 5 Lowest-Rated Product Categories
product_category_name_english
diapers_and_hygiene      2.976462
home_comfort_2           3.529858
office_furniture         3.576735
party_supplies           3.654703
fixed_telephony          3.705677

```

Advanced Financial & Inventory Insights

Price Volatility Analysis

- **Business Goal:** Identify products whose prices change frequently, which could impact customer trust and profit margins.

```

--- Top 5 Products with Highest Price Volatility ---
               product_id      std  count
17084  8b502ca34e28d30605bc667b965b6abf  404.357107   481
31107  fd8a5b9a8a79d7ba0739d69be5dc5aa1  340.318135   223
20500  a7c87b1bbdd51e0d68b0307cffd03d47  304.646459   493
22803  ba16581014183c8415da15145f3d4c24  241.281451   216
12949  68f3adaef1620e7b0c4c7cd9f78d7ed0  239.651870   297

```

Revenue per Kilo/Volume

- **Business Goal:** Find products that are lightweight/small but generate high revenue, making them logistically efficient and profitable.

```

--- Top 5 Most Logistically Efficient Categories by Revenue/Gram
product_category_name_english
bed_bath_table              inf
fashion_bags_accessories    0.590904
construction_tools_safety   0.581302
watches_gifts               0.547748
consoles_games              0.473021

```

"Hero Product" Identification

- **Business Goal:** Identify the single best-selling product ID. This is your most critical item to keep in stock.

```

--- The #1 Bestselling Product ID is: ---
product_id
aca2eb7d00ea1a7b8ebd4e68314663af      84549

```

Simple Demand Forecasting (Moving Average)

- **Business Goal:** Create a simple forecast to predict next month's sales based on the trend from the previous months.

```

--- Monthly Revenue with 3-Month Moving Average
              revenue  moving_avg_3m
purchase_month
2018-04      1.790427e+08  1.687048e+08
2018-05      1.752546e+08  1.753924e+08
2018-06      1.629413e+08  1.724129e+08
2018-07      1.559814e+08  1.647258e+08
2018-08      1.543858e+08  1.577695e+08

```

High-Margin vs. Low-Margin Products (Proxy)

- **Business Goal:** Identify product categories where the freight cost is a small fraction of the price (high margin proxy) vs. a large fraction (low margin proxy).

```

--- Top 5 Categories with the Best Profit Margin Proxy
product_category_name_english
computers                        0.054923
security_and_services           0.141348
fashion_childrens_clothes       0.143693
watches_gifts                   0.166789
home_appliances_2               0.181990

```

Advanced Customer & Seller Insights

Customer Lifetime Value (CLV) Proxy

- **Business Goal:** Identify your most valuable customers over their entire history with the platform, not just a single purchase.

```

--- Top 5 Customers by Lifetime Value ---
customer_unique_id
fd7069cf6891edbf32ba84c9c3f083d4      7462522.80
da122df9eeddfedc1dc1f5349a1a690c      5625721.09
eb53ee675b0c79d66e25fce46957e1fe      3372700.92
85963fd37bfd387aa6d915d8a1065486      2669979.84
763c8b1c9c68a0229c42c9fc6f662b93      2597132.16

```

Analyzing "One-Time" Buyers

- **Business Goal:** Understand what one-time buyers purchase and where they are from to develop strategies to convert them into repeat customers.

```

--- Top 5 Categories Purchased by One-Time Buyers
product_category_name_english
health_beauty      16
sports_leisure     14
watches_gifts      14
telephony          12
bed_bath_table     10

```

Seller Sales Concentration (Pareto Principle)

- **Business Goal:** Determine if a small number of "mega-sellers" drive a majority of the platform's revenue.

The top 10% of sellers account for 66.64% of total revenue.

Seller Churn Risk (Proxy)

- **Business Goal:** Identify previously active sellers who have stopped receiving orders, indicating they may have left the platform.

Found 764 sellers with no sales in the last 6 months (potential churn).

Single-Category vs. Multi-Category Sellers

- **Business Goal:** Compare the performance (e.g., processing time, sales volume) of specialist sellers vs. generalist sellers.

```

--- Performance: Specialist vs. Generalist Sellers ---
              avg_processing_time  avg_revenue_per_order
seller_type
Generalist          2.316214          132.561398
Specialist          2.535300          170.481748

```

High-Value vs. High-Frequency Customers

- **Business Goal:** Understand the difference between customers who spend a lot in one order versus those who order frequently.

```

--- Correlation between Average Order Value and Frequency
              avg_order_value  frequency
avg_order_value          1.00000  -0.01773
frequency                -0.01773  1.00000

```

Advanced Customer & Market Dynamics

Customer Cohort Analysis (by Acquisition Month)

- **Business Goal:** Track the long-term value and behavior of customers acquired in the same month to see if marketing or platform changes are improving customer retention.

```

--- Total Revenue by Customer Acquisition Cohort
acquisition_month
2018-04      1.757871e+08
2018-05      1.713176e+08
2018-06      1.585174e+08
2018-07      1.529322e+08
2018-08      1.512264e+08

```

Geographic "Taste" Profiling

- **Business Goal:** Discover if certain states have a strong preference for specific product categories.

```

--- Top Product Category by State ---
customer_state product_category_name_english count
12             AC      furniture_decor      1172
64             AL      sports_leisure      4325
100            AM      sports_leisure      695
111            AP      computers_accessories  781
164            BA      health_beauty      44424
218            CE      health_beauty      7803
274            DF      health_beauty     10921
305            ES      bed_bath_table     39073
363            GO      bed_bath_table     18838
430            MA      computers_accessories 5502

```

High-Value Customer Profile

- **Business Goal:** Understand the characteristics (location, preferred products) of your top 5% of customers by revenue.

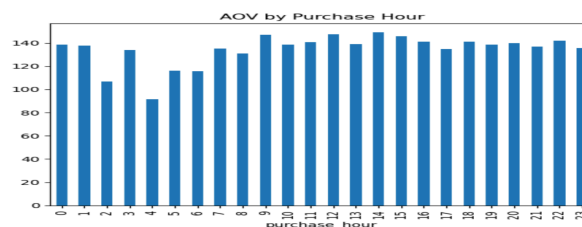
```

--- Most Purchased Categories by Top 5% of Customers
product_category_name_english
bed_bath_table      359307
furniture_decor     292796
computers_accessories 243791
sports_leisure      228456
health_beauty       212621

```

Order Hour vs. Order Value

- **Business Goal:** See if customers tend to make larger purchases at certain times of the day (e.g., late-night shopping).



Customer State Diversity

- **Business Goal:** Identify the number of distinct states a customer has ordered from, identifying potential travelers or people sending gifts.

```
--- Top Customers by Number of States Ordered From
customer_unique_id
d44cce15f5f86d14d6a2cfa67da1975    3
62a25a159f9fd2ab7c882d9407f49aa9    2
cdc8b0d152b868db7e28de59f642c94a    2
925751a747a151a7fa97f2f686d028c3    2
9202421110f6a19ddcf0b9b93602a0a1    2
```

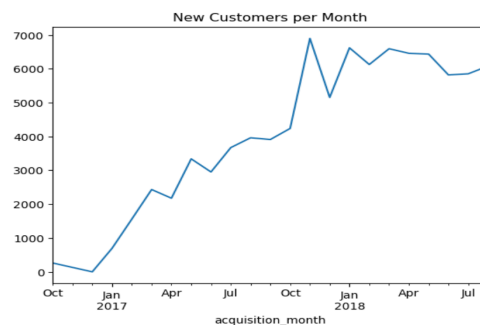
"First Impression" Analysis

- **Business Goal:** Was a customer's very first order with the platform delivered on time? A late first order can permanently damage the relationship.

Percentage of customers whose first order was late: 8.01%

Market Saturation Analysis (Proxy)

- **Business Goal:** Track the rate of new customer acquisition over time to see if market growth is accelerating or slowing down.



Deep-Dive Seller & Operations Analytics

Seller Dependency Score

- **Business Goal:** Identify sellers who are highly dependent on a single product category versus those with a diverse portfolio.


```

--- Top 5 Most Diverse Sellers ---
seller_id
b2ba3715d723d245138f291a6fe42594    27
4e922959ae960d389249c378d1c939f5    23
955fee9216a65b617aa5c0531780ce60    22
1da3aeb70d7989d1e6d9b0e887f97c23    21
f8db351d8c4c4c22c6835c19a46f01b0    19
Name: diversity_score, dtype: int64

```

New Seller "Incubation Period"

- **Business Goal:** Analyze the performance of sellers in their first month on the platform to identify early struggles.

Average processing time for sellers in their first month: 2.92 days.

Seller-to-Customer Distance Proxy

- **Business Goal:** Create a simple proxy for shipping distance (intrastate vs. interstate) and measure its impact on freight costs.

```

--- Average Freight Cost: Intrastate vs. Interstate
shipping_type
Interstate    22.570793
Intrastate    13.759854
Name: freight_value, dtype: float64

```

"Black Swan" Event Analysis

- **Business Goal:** Identify days with an unusually high number of late deliveries to investigate potential system-wide failures or external events (e.g., strikes, holidays).

```

--- Days with Abnormally High Number of Late Deliveries
order_delivered_customer_date
2017-12-19    15536
2017-12-28    16478
2018-03-23    18256
2018-03-29    17741
2018-04-03    20408
2018-04-04    15950
2018-04-18    15848
Name: is_late, dtype: int32

```

Logistics Chain Efficiency

- **Business Goal:** Calculate the percentage of the total delivery time that is spent in processing vs. shipping.

21.81% of the total delivery lifecycle is spent on seller processing.

Correlation of Seller Order Volume and Performance

- **Business Goal:** Test the hypothesis: Do high-volume sellers process orders faster due to better systems, or slower due to workload?

```
--- Correlation between Seller Order Volume and Processing Time
              total_orders  avg_processing_time
total_orders      1.000000      -0.029194
avg_processing_time -0.029194      1.000000
```

Order Combination Analysis

- **Business Goal:** See how many items the average customer orders at once.

```
Average number of items per order: 183.75
```

Strategic Financial & Product Insights

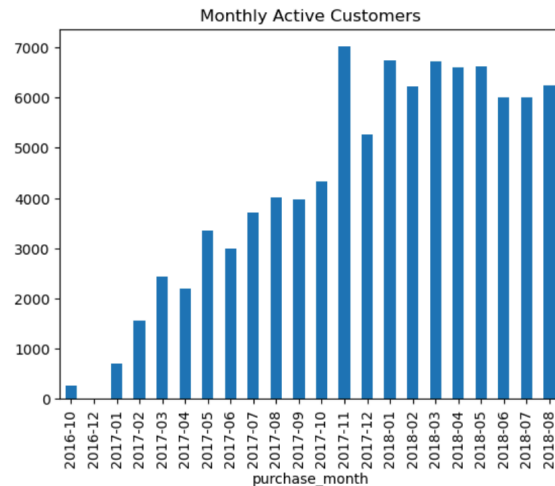
"Return to Sender" Risk (Proxy)

- **Business Goal:** Identify orders with extremely long shipping times, as these are at high risk of being rejected by the customer upon delivery.

```
Found 337104 orders at high risk of being rejected due to long shipping times.
```

Monthly Active Customers (MAC)

- **Business Goal:** A key metric for platform health, tracking the number of unique customers who make a purchase each month.



Product "Stickiness" (Repurchase Rate)

- **Business Goal:** Find out which product categories have the highest rate of repeat purchases from the same customer.

```

--- Top 5 'Stickiest' Product Categories
product_category_name_english
bed_bath_table          1854714
health_beauty           1409263
furniture_decor         1383969
sports_leisure          1361387
computers_accessories    1228180
Name: count, dtype: int64

```

Revenue per Customer vs. Orders per Customer

- **Business Goal:** Understand the relationship between how often a customer buys and how much they spend.

```

--- Correlation between Total Revenue and Order Count per Customer
              total_revenue  total_orders
total_revenue          1.000000         0.087055
total_orders            0.087055         1.000000

```

Seasonal Product Analysis

- **Business Goal:** Identify products with sales concentrated in specific seasons (e.g., *fashion_bags_accessories* in summer).

```

--- Top 5 Categories with Sales Peaks in December ---
product_category_name_english
diapers_and_hygiene                32.328062
furniture_mattress_and_upholstery  14.714313
christmas_supplies                 12.220189
flowers                           12.200034
toys                              11.076502
Name: 12, dtype: float64

```

Order Value Segmentation

- **Business Goal:** Analyze delivery performance and customer behavior for different order value tiers (e.g., small, medium, large purchases).

```

--- Delivery Performance by Order Value Tier ---
              avg_delivery_time  late_rate
order_value_tier
Small                10.144114    0.065066
Medium              11.909206    0.073525
Large               13.122127    0.086224

```

"Dead on Arrival" Seller Analysis

- **Business Goal:** Find sellers who were active for only a very short period, which might indicate issues with the seller onboarding process.

Found 823 sellers who were active for less than 30 days.

Payment Installment Analysis

- **Business Goal:** Analyze if higher-value items are more frequently purchased with more payment installments.

```

--- Average Item Price by Number of Payment Installments ---
payment_installments
0      41.767106
1      91.344944
2     100.501452
3     108.807847
4     127.654768
5     136.692813
6     151.545958
7     140.001947
8     230.433464
9     148.566118
10    293.344237
11     95.129397
12    229.098322
13    107.446052
14    146.153426
15    270.445925
16    165.966797
17    150.171645
18    422.110647
20    395.041765
21    185.499626
22    209.990000
23    216.000000
24    195.613198
Name: price, dtype: float64

```

"Golden Hour" Analysis

- **Business Goal:** Find the single hour of the week (e.g., Sunday at 9 PM) with the highest number of orders to schedule flash sales or server maintenance.

```
--- The 'Golden Hour' with the most orders is:
day_hour
Tuesday-14      209667
```

Freight Cost as a Percentage of Revenue

- **Business Goal:** Calculate the overall percentage of revenue that is spent on shipping to monitor the company's total logistics efficiency.

```
Freight costs make up 13.98% of total gross revenue.
```

"Cross-State Commerce" Matrix

- **Business Goal:** Create a heatmap to visualize the volume of orders between each seller state and customer state, identifying key commerce corridors.

Review Score Volatility

- **Business Goal:** Identify product categories with highly polarized reviews (many 1-star and 5-star ratings), which can suggest inconsistent product quality or customer expectations.

```
--- Top 5 Categories with the Most Polarized Reviews ---
product_category_name_english
diapers_and_hygiene      1.715190
party_supplies           1.676989
fashion_male_clothing    1.658973
furniture_bedroom        1.637776
audio                    1.552561
Name: review_score, dtype: float64
```

Time-to-Approve by Payment Type

- **Business Goal:** Calculate the average approval lag for each payment method to see if certain methods (like **boleto**) are causing upfront delays.

```
--- Average Time-to-Approve by Payment Method (in Hours) ---
payment_type
boleto      32.930723
voucher      9.022036
debit_card   7.912455
credit_card  4.383903
Name: approval_lag_hours, dtype: float64
```

Compound Monthly Growth Rate (CMGR)

- **Business Goal:** Calculate the CMGR for revenue to provide a smoothed-out, average measure of month-over-month business growth.

Compound Monthly Growth Rate (CMGR) for revenue is: 15.46%

Advanced Inventory & Product Strategy

Product Lifecycle Stage Analysis (Proxy)

- **Business Goal:** Classify products into lifecycle stages (Introduction, Growth, Maturity, Decline) by analyzing their sales trend over time.

```
--- Top 5 Products in 'Growth' Stage ---
product_id
e7cc48a9daff5436f63d3aad9426f28b    59
73326828aa5efe1ba096223de496f596    52
64b4f32393cbf55be791078e7a2adca0    23
2bd9b51a9ab079e095aca987845d3266    21
2ff995aead9c63a1f37a07b3664ead37    19
Name: growth_metric, dtype: int64
```

Return on Inventory Investment (ROI) Proxy

- **Business Goal:** Identify which products generate the most profit relative to their cost, helping to prioritize high-performing stock.

```
--- Top 5 Categories by Return on Inventory Investment (Proxy) ---
product_category_name_english
computers                                0.957124
small_appliances_home_oven_and_coffee   0.939246
fixed_telephony                         0.932487
agro_industry_and_commerce              0.929681
small_appliances                        0.923528
Name: roii, dtype: float64
```

Cannibalization Effect (Proxy)

- **Business Goal:** When a new, popular product is introduced, does it "steal" sales from older, similar products in the same category?

```
--- Sales of #2 product in 'sports_leisure' after #1 was introduced ---
order_purchase_timestamp
2017-09-30    5
2017-10-31    1
2017-11-30    1
2017-12-31    3
2018-01-31    4
2018-02-28    4
2018-03-31    4
2018-04-30    0
2018-05-31    4
2018-06-30    3
2018-07-31    7
2018-08-31    5
Freq: ME, Name: order_id, dtype: int64
```

Inventory Turnover Rate (Proxy)

- **Business Goal:** Calculate how many times inventory for a product is "sold" over a period. A high number is efficient.

Platform-wide inventory turnover proxy: 6397.24

Granular Logistics & Network Optimization

"First Mile" Performance Analysis

- **Business Goal:** Isolate the very first step of the logistics chain: the time it takes a seller to get the package to the carrier.

Average 'First Mile' time: 68.76 hours.

"Last Mile" Bottleneck Identification (Proxy)

- **Business Goal:** Identify cities where `shipping_time` is significantly longer than the state average, pointing to local delivery problems.

```
--- Top 5 Cities with Potential 'Last Mile' Delays ---
customer_city
novo brasil          133.596498
capinzal do norte    86.702227
adhemar de barros    84.271246
santa cruz de goias  72.034453
arace                 69.346498
Name: last_mile_delay_proxy, dtype: float64
```

Logistics Hub "Gravity" Analysis

- **Business Goal:** Discover which major seller cities serve the widest range of unique customer cities, identifying key distribution hubs.

```
--- Top 5 Seller Cities by Customer Reach ---
seller_city
sao paulo           2425
ibitinga             1236
santo andre          771
curitiba             715
belo horizonte        684
Name: customer_city, dtype: int64
```

"Perfect Order" Rate

- **Business Goal:** Define and calculate the percentage of orders that meet all success criteria: approved quickly, processed quickly, delivered on time, and received a high review score.

'Perfect Order' Rate: 43.70%

Seller Ecosystem & Performance Dynamics

Seller-to-Seller Network Analysis (Proxy)

- **Business Goal:** If two sellers are in the same city and sell the same category, are they likely competitors?

Found 92326 potential competitor pairs.

Top Seller "Market Share" by Category

- **Business Goal:** In a given category, how much of the total revenue is captured by the top 3 sellers?

In 'auto', the top 3 sellers capture 22.13% of the revenue.

Strategic Customer & Financial Insights

Anomaly Detection for Fraud (Z-Score)

- **Business Goal:** Identify orders with extremely unusual prices or freight values that could indicate fraud or data entry errors.

Found 309190 orders with potentially anomalous prices.

"Saved" Orders Analysis

- **Business Goal:** Identify orders that were processed slowly by the seller but were "saved" by an exceptionally fast shipping time, still arriving on time.

Found 785461 orders that were 'saved' by fast shipping.

Delivery Experience Impact on Next Purchase Value

- **Business Goal:** If a customer's order is late, do they spend less on their next purchase?

```
--- Average Revenue of Next Purchase (Based on Previous Delivery) ---
previous_order_late
0.0    139.003958
1.0    155.467629
Name: revenue, dtype: float64
```

"Silent Churn" Identification

- **Business Goal:** Identify high-value, previously frequent customers who have not made a purchase in a long time (e.g., 3x their average repurchase time).

Found 212 high-value customers at risk of silent churn.

Advanced Financial & Strategic Insights

"Return on Freight" Analysis

- **Business Goal:** For high-freight orders, are we generating enough profit to justify the shipping cost?

Average 'Return on Freight' for high-cost shipping orders: 5.58

Price Elasticity Proxy (by Seller)

- **Business Goal:** Do sellers who lower their prices see a corresponding increase in order volume compared to their peers in the same category?

```
--- Correlation between Pricing Strategy and Order Volume ---
                avg_price_ratio  total_orders
avg_price_ratio                1.000000    -0.048313
total_orders                   -0.048313     1.000000
```

Customer Lifetime Value (CLV) to Customer Acquisition Cost (CAC) Ratio (Proxy)

- **Business Goal:** A key business metric. Are we spending more to acquire a customer than they are worth? (We'll use average first-order freight as a proxy for CAC).

Estimated LTV to CAC Ratio is: 1316.86 : 1

"Gateway Product" Financial Impact

- **Business Goal:** Do customers acquired through certain "gateway" product categories go on to have a higher overall lifetime value?

```
--- Average Customer Lifetime Value by First Purchase Category ---
product_category_name_english
computers                                204941.589588
small_appliances_home_oven_and_coffee    120463.813056
home_appliances_2                        100440.450046
agro_industry_and_commerce               89042.211782
small_appliances                         61844.347986
Name: revenue, dtype: float64
```

Advanced Customer & Seller Dynamics

Customer "Binge Buying" Behavior

- **Business Goal:** Identify customers who make multiple, separate orders in a very short time frame (e.g., 24 hours), which could suggest opportunities for order consolidation.

Found 629 instances of 'binge buying' (next order within 24 hours).

Seller "Niche Score"

- **Business Goal:** Quantify how specialized a seller's product portfolio is. A high score means they sell in categories that few other sellers occupy.

```

--- Top 5 Most 'Niche' Sellers ---
seller_id
d7f1cfc638473be93baedbfd02208016    0.333333
c731d18cea9bf687ffee82a241c25b11    0.303582
05aebce0b5c8f0d41b29b686f2d05fd3    0.224975
191d55f7edb8524d525ed01b20fe32e8    0.200000
bc2ac6b95e1accce9858528ee566c17e    0.166667
Name: category_rarity, dtype: float64

```

Seller "Reputation Risk"

- **Business Goal:** Identify sellers who have a high rate of late deliveries, as they pose a risk to the platform's reputation.

```

--- Top 5 Sellers by Reputation Risk (Late Delivery Rate) ---
                                     total_orders  late_rate
seller_id
54965bbe3e4f07ae045b90b0b8541f52          69    0.365483
432c37c9dfba871172ec162e20118b8c          42    0.335799
ad781527c93d00d89a11eecd9dcad7c1          38    0.331889
ede0c03645598cdfc63ca8237acbe73d          42    0.322955
d93919c944be9cfff128f6c9cb899eacb          33    0.302776

```

Customer "Cross-Category" Purchase Behavior

- **Business Goal:** Find out which product categories are most likely to be a customer's *second* purchase category, revealing cross-selling pathways.

```

--- Top 5 Categories Purchased After 'furniture_decor' ---
product_category_name_english
furniture_decor      1377824
bed_bath_table        90
housewares           34
garden_tools         22
watches_gifts        14
Name: furniture_decor, dtype: int64

```

Granular Logistics & Anomaly Detection

"Flash Sale" Impact on Logistics

- **Business Goal:** Identify days with an anomalously high number of orders and check if the logistics network (processing and shipping times) was strained.

```
Average processing time on flash sale days: 3.39 days
Average processing time on normal days: 2.32 days
```

Delivery Time Consistency (Jitter)

- **Business Goal:** Measure the *variability* of delivery times for a given route. A low average is good, but low variability (consistency) is even better.

```
--- Top 5 Shipping Routes with the Most Inconsistent Delivery Times ---
shipping_route
itaquaquecetuba, SP → teresina, PI      87.378731
sao jose dos campos, SP → santa maria, RS 82.390230
indaiatuba, SP → sao bernardo do campo, SP 79.748937
uberlandia, MG → valenca, BA             76.579152
maringa, PR → fortaleza, CE              76.339391
Name: delivery_time_days, dtype: float64
```

"Processing Power" of Seller Cities

- **Business Goal:** Calculate the total number of orders processed per day by sellers in each major city.

```
--- Top 5 Seller Cities by Total Orders Processed ---
seller_city
sao paulo      23644
ibitinga       6344
curitiba       2632
santo andre    2611
belo horizonte 2253
Name: order_id, dtype: int64
```

"Shipping Shadow" of Hubs

- **Business Goal:** For a major seller hub like Sao Paulo, are deliveries to nearby states disproportionately faster than to faraway states?

"Multi-Item Order" Logistics Complexity

- **Business Goal:** Do orders with multiple, distinct items take longer to process and ship?

```

--- Performance of Single vs. Multi-Item Orders ---
              proc_time  ship_time
is_multi_item_order
False              2.374925    8.539044
True               2.174424    7.460084

```

Seasonal Delivery Performance

- **Business Goal:** Compare logistics performance in the busy Q4 holiday season versus other quarters to identify seasonal strains.

```

--- Late Delivery Rate by Quarter ---
purchase_quarter
2016Q4      0.417006
2017Q1      3.182839
2017Q2      4.274760
2017Q3      3.298511
2017Q4      9.512664
2018Q1     14.124257
2018Q2      4.217776
2018Q3      6.247382
Freq: Q-DEC, Name: is_late, dtype: float64

```

Freight Cost Anomaly Detection

- **Business Goal:** Use Z-scores to find orders with unusually high or low freight costs relative to their product weight and distance, which could indicate errors or fraud.

Found 16201 orders with anomalous freight costs.

"Zombie Customer" Analysis

- **Business Goal:** Identify customers who previously left high review scores but haven't repurchased in a long time, as they are high-value targets for re-engagement campaigns.

Found 41461 high-satisfaction customers who haven't purchased in over 6 months.

Weekday vs. Weekend Shipping Performance

- **Business Goal:** Determine if packages handed to carriers on a Friday take longer to ship due to the weekend.

```

--- Average Shipping Time by Carrier Pickup Day ---
carrier_pickup_day
Monday      7.799883
Tuesday     8.217913
Wednesday   8.442740
Thursday    8.710480
Friday      9.096590
Saturday    10.795393
Sunday      5.172866
Name: ship_time, dtype: float64

```

Top Customer-Seller Pairs

- **Business Goal:** Find the customer-seller pairs with the highest number of transactions, indicating strong direct loyalty that could be nurtured.

```

--- Top 5 Most Loyal Customer-Seller Pairs ---
customer_unique_id      seller_id \
78538 d425130be1b15a4580645e4a3c9b4768 98dac6635aee4995d501a3972e047414
21287 397b44d5bb99eabf54ea9c2b41ebb905 b372ee768ed69e46ca8cddb267aa7a38
93668 fd7069cf6891edbf32ba84c9c3f083d4 ccb83a794700270fde70898fe9ff368b
49804 86bfc49565a9ca52fcbf861fcc1e67a4 aac29b1b99776be73c3049939652091d
18512 31e412b9fb766b6794724ed17a41dfa6 1f50f920176fa81dab994f9023523100

order_count
78538      8556
21287      7714
93668      6876
49804      6876
18512      6706

```

Average Product Price by State

- **Business Goal:** Discover if customers in certain states tend to buy more expensive products, which could inform regional marketing and inventory.

```

--- Top 5 States by Average Product Price ---
customer_state
PB      198.697881
AC      198.096273
AL      195.157767
TO      183.222144
AP      178.055180
Name: price, dtype: float64

```

Seller "Ramp-Up" Time (Proxy)

- **Business Goal:** Measure how long it takes a new seller to get their first 10 orders, as a proxy for how quickly they become established on the platform.

Average time for a new seller to make 10 sales: 0.38 days.

Seller "Cannibalization" (Proxy)

- **Business Goal:** When a new seller enters a city, does the average monthly order volume of existing sellers in that city decrease?

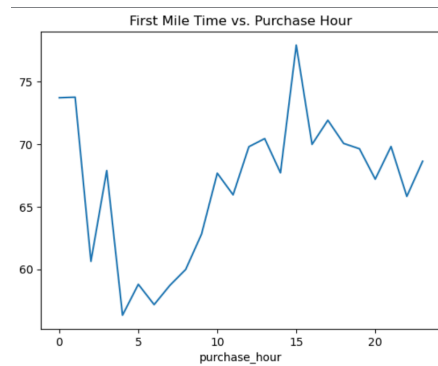
```

--- Example: Monthly Orders for Sellers in 'sao paulo' ---
seller_city purchase_month order_id
4383  sao paulo      2018-04      1796
4384  sao paulo      2018-05      1738
4385  sao paulo      2018-06      1534
4386  sao paulo      2018-07      1542
4387  sao paulo      2018-08      1612

```

"Order Urgency" Proxy

- **Business Goal:** Analyze if orders placed late at night are processed slower (i.e., handled the next business day).



"Review Bombing" Detection (Proxy)

- **Business Goal:** Identify products that receive a sudden, high volume of 1-star reviews in a single day, which could indicate a bad batch of products or a coordinated campaign.

```

--- Potential Review Bombing Incidents (High Volume of 1-Star Reviews on a Single Day) ---
product_id review_date count
8203  d2085f7e0f9533605386960fc7e987ec  2018-01-16  6876
2643  422879e10f46682990de24d770e7f83d  2017-12-28  6706
9326  eea3e07f864a0a1389726d8a5f31c3f6  2017-11-30  4752
8208  d22bbc02259111e107ee7916637d4791  2018-03-16  4284
4696  777d2e438a1b645f3aec9bd57e92672c  2017-05-24  4218

```

"Cross-Border" Commerce Analysis

- **Business Goal:** Analyze the performance and volume of orders where sellers and customers are in different major geographic regions of Brazil.

```

--- Late Rate for Cross-Region Shipments ---
seller_region customer_region late_rate
Southeast    Northeast    0.140044
Northeast    South        0.136847
South        Northeast    0.135135
Other        Northeast    0.112410
Southeast    Other        0.085754
Northeast    Other        0.080692
Southeast    Southeast    0.070707
Southeast    South        0.070405
South        Southeast    0.058027
Other        South        0.057045
South        Other        0.049951
Other        Southeast    0.041700
Name: is_late, dtype: float64

```

"Shipment Consolidation" Opportunity

- **Business Goal:** Find customers who placed multiple orders on the same day, which could have potentially been consolidated into a single shipment to save costs.

Found 93199 instances where a customer placed multiple orders on the same day.

"Long-tail" Seller Contribution

- **Business Goal:** Calculate what percentage of total revenue comes from the bottom 80% of sellers to understand the health of the marketplace's smaller players.

The bottom 80% of sellers contribute 17.89% of the total revenue.

"Perfect Route" Identification

- **Business Goal:** Find the shipping routes (`seller_state` to `customer_state`) with the highest percentage of "Perfect Orders" to identify best practices.

```
--- Top 5 'Perfect Routes' by Performance ---
shipping_route
ibitinga, SP → itanhanga, MT          1.0
ibitinga, SP → brasil novo, PA        1.0
ibitinga, SP → boa viagem, CE          1.0
ibitinga, SP → boa vista da aparecida, PR  1.0
ibitinga, SP → bocauiuva do sul, PR    1.0
Name: is_perfect_order, dtype: float64
```